

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Precision (Level 3)

Total Marks: 30

Question Count: 3

Code of Conduct

I G.Jaahnavi(192525277) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Jaahnavi

Assessment Tasks

Case Study

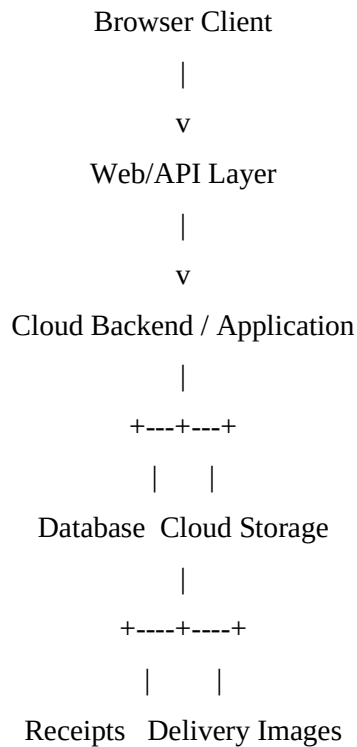
- CASE 1: A FOOD DELIVERY COMPANY WANTS TO DEPLOY A CLOUD-NATIVE BACKEND WITH WEB APIS, BROWSER CLIENTS, AND SHARED CLOUD STORAGE FOR RECEIPTS AND DELIVERY IMAGES. PROPOSE THE ARCHITECTURE AND EXPLAIN THE ROLE OF EACH COMPONENT.**

Case 1: Cloud-Native Food Delivery Backend

1.Problem Statement

A food delivery company needs a cloud-native backend that supports web APIs, browser clients, and shared cloud storage for receipts and delivery images.

2.Proposed Architecture



3.Components and Roles

- **Browser Client**
Provides the user interface.
Customers can place orders and track deliveries.
Communicates with the backend through web APIs.
- **Web APIs**
Connect the browser application with the cloud backend.
Handle requests such as placing orders, checking status, and retrieving information.
Provide controlled access to backend services.
- **Cloud Backend**
Processes business logic.
Manages orders, customers, payments, and delivery information.
Can scale according to the number of users.
- **Database**
Stores structured information such as customer details, order details, and delivery status.
Allows the application to retrieve and update information efficiently.
- **Cloud Object Storage**
Stores receipts, delivery images, and other files.
Suitable for large amounts of unstructured data.
Provides scalable storage and easy access.

4.Working

- Customer opens the browser application.
- Browser sends a request to the web API.
- The cloud backend processes the request.
- Order information is stored in the database.
- Receipts and delivery images are stored in cloud storage.
- The backend retrieves the required information and sends the response to the browser.

5.Advantages

- High scalability
- Centralized storage
- Easy access to files
- Better availability
- Supports many users
- Reduces the need for physical infrastructure

6.Conclusion

A cloud-native architecture using browser clients, web APIs, cloud backend services, database storage, and cloud object storage provides a scalable and practical solution for the food delivery company.

2. CASE 2: A MEDIA STARTUP MUST DECIDE BETWEEN SAN-BACKED STORAGE, OBJECT STORAGE, AND SERVERLESS PROCESSING FOR VIDEO UPLOAD WORKFLOWS. ANALYZE THE CASE AND RECOMMEND A DEPLOYMENT MODEL.

Case 2: Media Startup Video Upload Workflow

1.Problem Statement

A media startup must choose between SAN-backed storage, object storage, and serverless processing for its video upload workflow.

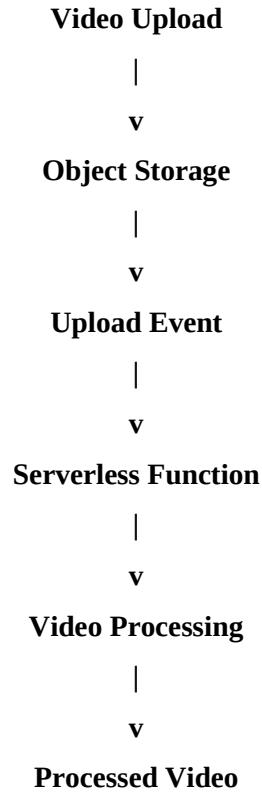
2.Analysis

Storage/Service	Main Use	Advantage	Limitation
SAN Storage	High-performance block storage	High performance	More complex and costly
Object Storage	Video and large files	Highly scalable and durable	Not designed like traditional block storage
Serverless Processing	Automatic video processing	No server management	Depends on cloud service limits

3.Recommended Model

The best approach is a combination of object storage and serverless processing.

Proposed Workflow



4.Working

- User uploads a video.
- The original video is stored in object storage.
- An upload event triggers serverless processing.
- The serverless service processes the video.
- The processed video is stored back in object storage.
- Users can access the processed video when required.

5.Why Object Storage?

- Suitable for large video files.
- Highly scalable.
- Can store huge numbers of files.
- Provides centralized access.
- Reduces the need to manage storage servers.

6.Why Serverless?

- Automatically runs processing when needed.
- No need to continuously maintain servers.
- Can scale according to workload.
- Suitable for event-based video processing.

7.Role of SAN

SAN can provide high performance, but for a cloud-based media startup it may introduce additional management and infrastructure requirements.

8.Conclusion

For this case, object storage + serverless processing is the recommended deployment model because it provides scalability, flexible storage, and automatic processing with less infrastructure management.

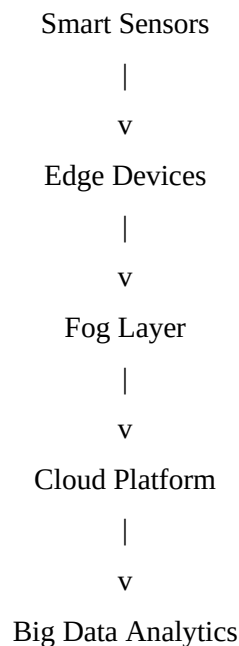
Case 3: A SMART CITY PLATFORM NEEDS LOW-LATENCY DECISION MAKING NEAR SENSORS WHILE STILL USING THE CLOUD FOR ANALYTICS. ANALYZE THE USE OF EDGE COMPUTING AND FOG COMPUTING IN THIS ENVIRONMENT.

Smart City Edge and Fog Computing

1.Problem Statement

A smart city platform needs low-latency decision making near sensors while still using the cloud for analytics. The assessment specifically asks for analysis of edge and fog computing in this environment.

2.Proposed Architecture



3.Edge Computing

Edge computing processes data close to where it is generated.

Examples:

- Traffic sensors
- CCTV cameras
- Smart parking sensors
- Environmental sensors

4.Benefits of Edge Computing

- Very low latency.
- Faster decisions.
- Reduces unnecessary data transmission.
- Can continue some processing even with limited cloud connectivity.
- Useful for real-time applications.

5.Fog Computing

Fog computing provides an intermediate processing layer between edge devices and the cloud.

The fog layer can:

- Collect data from multiple sensors.
- Filter and aggregate data.
- Perform local analysis.
- Forward important information to the cloud.

6.working Example

- Traffic sensors collect vehicle information.
- Edge devices process immediate traffic conditions.
- Fog nodes combine information from multiple nearby sensors.
- A traffic decision can be made quickly.
- Important summarized data is sent to the cloud.
- The cloud performs large-scale analytics and long-term analysis.

7.Edge vs Fog vs Cloud

Feature	Edge	Fog	Cloud
Location	Near sensor	Between edge and cloud	Centralized
Latency	Very low	Low	Higher
Processing	Local	Regional	Large-scale
Main purpose	Immediate decisions	Local coordination	Analytics/storage
Example	Traffic sensor	City gateway	Cloud analytics

8.Advantages

- Faster response
- Reduced network traffic
- Efficient resource usage
- Real-time decision making
- Cloud can handle large-scale analytics

9.Conclusion

A combined Edge–Fog–Cloud architecture is suitable for the smart city. Edge devices handle immediate decisions, fog nodes coordinate and filter local data, and the cloud performs large-scale analytics and storage. This provides both low latency and powerful cloud-based analysis.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized